TerraFlow BioAI: A Zone-Aware Edge-IoT Architecture for Closed-Loop Precision Irrigation and Multi-Parametric Agro-Advisory

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Abstract—Agricultural irrigation accounts for more than 70% of global freshwater withdrawals, yet the dominant control methods in developing regions—fixed-timer schedules and uncalibrated threshold switching—remain fundamentally open-loop. They ignore soil hydraulic properties, crop phenology, and real-time volumetric feedback, leading to chronic over-irrigation and yield-limiting water stress during critical growth stages.

This paper introduces TerraFlow BioAI, a zone-aware edge-IoT platform that closes the irrigation loop at the hardware level. A single field node (TF01) manages one physical zone (Z01) through a dual-controller topology: an STM32F103C8T6 microcontroller serves as the deterministic local safety authority—running a 1-D Kalman filter on the soil-moisture signal, enforcing a five-state finite-state machine, and terminating pump actuation based on measured pulse-flow volume—while an ESP32 handles upstream JSON telemetry over Wi-Fi. The agronomic decision framework draws on an 80-row knowledge base derived from FAO-56 and FAO-33 standards, covering four Indian soil classes and four crops across five phenological stages. A Random Forest model provides decision support without overriding hardware safety gates.

We present the complete architecture, the governing equations, and benchtop verification results from the completed prototype (Phases 1–2). Because multi-week comparative field trials (Phases 3–4) are still in progress, we formally define the experimental protocol and evaluation metrics rather than reporting premature findings.

Index Terms—Precision agriculture, closed-loop irrigation, edge computing, IoT, Kalman filter, FAO-56, cyber-physical systems, STM32

This work was carried out at the Department of Electronics and Telecommunication Engineering, Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology (SPIT), Mumbai, India.

I. Introduction

Agricultural water management sits at the intersection of two realities. On one hand, India alone uses more than 80% of its renewable freshwater for irrigation [1]; on the other, the majority of farms still rely on fixed-schedule watering or single-threshold IoT switches that treat every crop, every soil, and every growth stage identically [2]. The consequences—waterlogging during monsoon-adjacent stages, stress-induced sterility during flowering, and silent pump failures during unattended operation—are well documented yet rarely addressed at the embedded-system design level [3].

The recent Artificial Intelligence of Things (AIoT) literature has introduced cloud-centric platforms that stream raw telemetry to remote servers where deep-learning models predict irrigation schedules and relay actuation commands back to the field [4], [6]. While computationally flexible, this paradigm introduces a critical safety hazard: a lost Wi-Fi packet during active pumping can leave the pump running indefinitely, draining the reservoir and flooding the root zone. Moreover, black-box neural networks offer no agronomic justification for their decisions, limiting farmer trust and academic reproducibility [3].

At the same time, the agronomic community has spent decades formalizing crop water requirements through the FAO-56 Penman-Monteith methodology [7] and the FAO-33 yield-response framework [8], yet these models are almost never integrated into embedded firmware. IoT

developers commonly hard-code moisture thresholds such as "pump ON if $< 30\%$" without reference to soil texture, root-zone depth, or growth-stage sensitivity.

TerraFlow BioAI bridges this disconnect through four design principles that, taken together, distinguish it from prior work:

1) Deterministic edge safety. The STM32 retains sole pump authority; the cloud may advise but never actuate. A hardware watchdog, a float-switch interlock, and a maximum-runtime guard ensure that the pump defaults to OFF under any fault condition.
2) Calibrated signal conditioning. A 1-D Kalman filter preprocesses the capacitive moisture signal on-chip, replacing the common but lag-prone rolling average with a recursive estimator that tracks the slow diffusion dynamics of soil water.
3) Agronomic volumetric control. Irrigation depth is computed from FAO-56 crop coefficients (K_c), soil-specific depletion fractions (p), and zone area. Pump shutoff is governed by pulse-flow integration rather than by a fixed timer.
4) Explainable reason codes. Every decision is tagged with a human-readable label (RUN_MOISTURE_LOW, SKIP_RAIN_CONTEXT, FAULT_TANK_LOW) that can be reviewed in the dashboard and audited after the fact.

The remainder of this paper is organized as follows. Section II reviews related work and identifies the technical gap. Section III presents the system architecture. Section IV formalizes the decision engines and governing equations. Section V reports prototype implementation and benchtop verification. Section VI defines the planned field-trial protocol. Section VII discusses limitations and future directions, and Section VIII concludes.

II. Related Work and Technical Gap

Early smart-irrigation prototypes coupled 8-bit AVR or PIC microcontrollers with resistive soil probes and GSM modules [1]. Resistive probes, however, suffer rapid galvanic corrosion when buried in moist soil, causing calibration to drift within weeks [5]. Modern designs therefore prefer capacitive probes that sense dielectric permittivity without direct metal-electrolyte contact.

Cloud-based frameworks on ThingsBoard, Blynk, AWS IoT, or Firebase have been widely reported [2], [6]. In the typical topology, field nodes stream raw ADC readings to a cloud VM, which runs an ANN, LSTM, or SVR model and returns an actuation command. Two weaknesses emerge: (i) network latency creates a safety window in which the pump state is undefined, and (ii) complex models provide no agronomic explanation for their output [3]. Several authors have proposed edge-AI alternatives [5], but most still rely on uncalibrated percentage thresholds rather than on validated soil-water models.

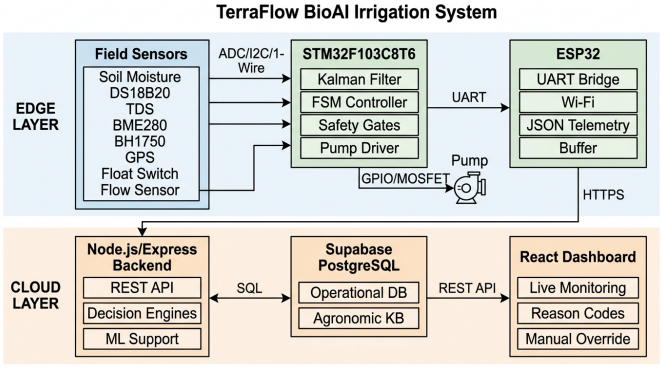


Fig. 1. TerraFlow BioAI system architecture. The edge layer (STM32 + ESP32) operates independently of the cloud for all safety-critical functions; the cloud layer provides analytics, advisory, and visualization.

TerraFlow addresses these gaps by embedding both the safety state machine and the calibrated threshold logic at the MCU layer, and by grounding every numerical threshold in FAO-56/FAO-33 published parameters rather than in arbitrary training-set statistics.

III. System Architecture

The architecture is organized into five cooperating layers, illustrated in Fig. 1. A single node TF01 manages zone Z01.

A. Layer 1: Field Instrumentation

Node TF01 employs six sensor modalities:

- Capacitive soil moisture (analog, 0–3.3 V).
- DS18B20 soil temperature (1-Wire, $\pm 0.5\,^{\circ}$C).
- TDS water-quality probe (0–1000 ppm). Note: TDS is not an NPK sensor; it provides irrigation-water salinity context only.
- BME280 (I2C): air temperature, relative humidity, barometric pressure.
- BH1750 (I2C): ambient illuminance (1–65 535 lux).
- NEO-6M GPS: zone geolocation for weather-API binding.

Two safety-critical inputs complete the sensing chain: a float switch (tank-level interlock) and a YF-S201 Hall-effect flow sensor (in-line volumetric metering).

B. Layer 2: Local Safety Authority (STM32)

The STM32F103C8T6 (Cortex-M3, 72 MHz, 64 KB Flash) executes bare-metal C++ firmware organized as a deterministic five-state machine:

- IDLE: Low-power standby between polling intervals.
- SENSE: Synchronous multi-channel acquisition, range validation, and Kalman filtering of the moisture channel.
- EVALUATE: Threshold comparison against cached agronomic setpoints and safety-interlock checks.

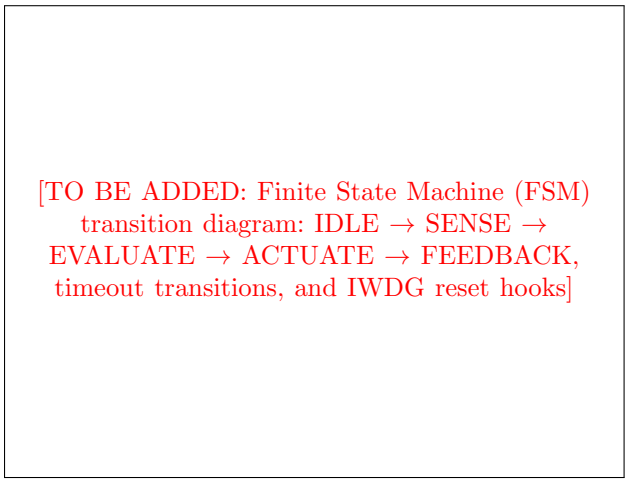


Fig. 2. Deterministic five-state execution machine running on STM32 edge authority. [TO BE ADDED: Replace with formal vector diagram or TikZ state chart.]

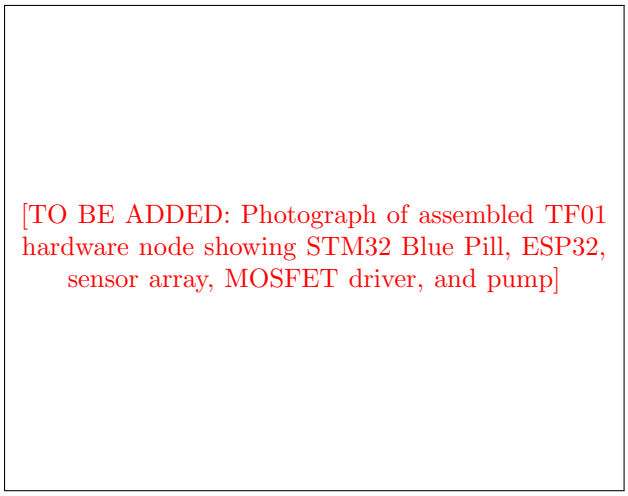


Fig. 3. Physical assembly of the TerraFlow edge node TF01. [TO BE ADDED: Replace with actual photo.]

- ACTUATE: MOSFET gate drive, pulse-flow interrupt counting, cumulative volume tracking toward V_{target}.
- FEEDBACK: Post-irrigation settling delay, moisture re-read, event logging.

The Independent Watchdog Timer (IWDG) resets the MCU if any state exceeds its allotted time window. A 10 kΩ gate pull-down ensures the MOSFET defaults to OFF during power-on, reset, or watchdog events.

C. Layer 3: Signal Conditioning (Kalman Filter)

Capacitive moisture readings are noisy due to ADC quantization, supply ripple, and soil heterogeneity at the probe tip. A 1-D Kalman filter, detailed in Section IV-A, replaces the common rolling average to provide a smoother, lower-latency estimate of the true moisture state.

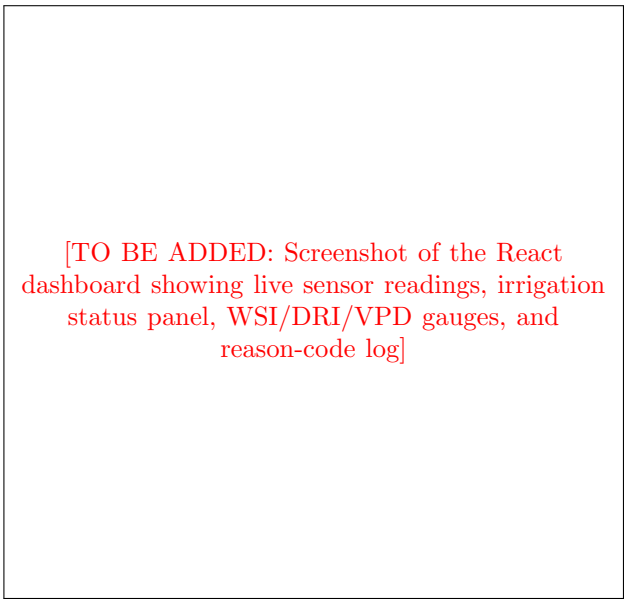


Fig. 4. TerraFlow React dashboard. [TO BE ADDED: Replace with actual screenshot.]

D. Layer 4: Connectivity Bridge (ESP32)

The ESP32 receives processed telemetry from the STM32 over UART at 115 200 baud, formatted as JSON packets with a CRC field and sequence counter. It manages Wi-Fi association, NTP synchronization, and HTTPS POST to the backend. When connectivity drops, the ESP32 caches packets in a local ring buffer and resumes transmission on reconnection. Critically, the ESP32 has no GPIO path to the pump; it is a communication pipe, not a safety authority.

E. Layer 5: Cloud Backend, Knowledge Base, and Dashboard

- Node.js/Express backend: ingests telemetry, validates timestamps, runs the Random Forest decision-support model, and exposes REST endpoints.
- Supabase (PostgreSQL): stores operational time-series data alongside the agronomic knowledge base (soil profiles, crop stages, fertilizer rules).
- React + Vite dashboard: renders live sensor gauges, irrigation history, reason-code logs, and manual override controls.

IV. Decision Engines and Governing Equations

A. Kalman Filter for Moisture Signal

Let x_k denote the true soil-moisture state at time step k and z_k the raw 12-bit ADC sample. We model the moisture as a slowly evolving process corrupted by measurement noise:

$$x_k = x_{k-1} + w_k, \qquad w_k \sim \mathcal{N}(0,\, Q), \tag{1}$$

$$z_k = x_k + v_k, \qquad v_k \sim \mathcal{N}(0,\, R), \tag{2}$$

TABLE I
Soil hydraulic reference parameters used in TerraFlow.

Soil class (Indian taxonomy)	AWC (mm/m)	Drainage quality	Freq. factor
Black (Vertisol, clay)	195	Poor	0.85
Red (Alfisol, sandy loam)	120	Moderate	1.35
Alluvial (Inceptisol, loam)	165	Moderate	1.00
Sandy (Arenosol, sand)	80	Excessive	1.60

where Q captures the variance of the physical moisture-diffusion process (set to 0.01 in our firmware) and R captures the sensor electrical noise (set to 2.5). These values were tuned empirically on the bench; field-specific tuning is planned.

At each step the filter predicts, then corrects:

$$\hat{x}_{k|k-1} = \hat{x}_{k-1|k-1}, \qquad P_{k|k-1} = P_{k-1|k-1} + Q, \tag{3}$$

$$K_k = \frac{P_{k|k-1}}{P_{k|k-1} + R}, \tag{4}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\big(z_k - \hat{x}_{k|k-1}\big), \tag{5}$$

$$P_{k|k} = (1 - K_k)\, P_{k|k-1}. \tag{6}$$

The filtered estimate $\hat{x}_{k|k}$ is the input to all downstream irrigation logic. This formulation is standard [11]; what is notable here is its deployment inside the MCU firmware as a preprocessing block that runs at the sensor polling rate.

B. Soil Hydraulic Parameters

The Available Water Capacity (AWC) defines the volumetric water held between field capacity θ_{FC} and permanent wilting point θ_{PWP}:

$$\mathrm{AWC} = 1000\,(\theta_{\mathrm{FC}} - \theta_{\mathrm{PWP}}) \quad [\mathrm{mm\,m^{-1}}]. \tag{7}$$

Table I lists the four Indian soil classes parameterized in the knowledge base, following FAO-56 Table 19 [7].

C. Sensor Calibration Model

Because the capacitive sensor outputs an analog voltage rather than a direct moisture percentage, a per-soil calibration maps the 12-bit ADC reading to the agronomic threshold. Under the convention higher ADC = drier soil, the pump-on threshold corresponding to a depletion fraction p is

$$\mathrm{ADC}_{\mathrm{on}} = \mathrm{ADC}_{\mathrm{FC}} + p\,\big(\mathrm{ADC}_{\mathrm{dry}} - \mathrm{ADC}_{\mathrm{FC}}\big), \tag{8}$$

and the pump-off threshold includes a hysteresis band h to prevent relay chatter:

$$\mathrm{ADC}_{\mathrm{off}} = \mathrm{ADC}_{\mathrm{FC}} - h. \tag{9}$$

Here $\mathrm{ADC}_{\mathrm{dry}}$ is measured on oven-dried or air-dried soil, and $\mathrm{ADC}_{\mathrm{FC}}$ on the same soil 24 hours after saturation.

[TO BE ADDED: Calibration curves for each soil type. Preliminary bench data available; field-calibrated curves pending.]

TABLE II
Crop-stage parameters from the agronomic knowledge base. DAT = days after transplanting; DAS = days after sowing.

Crop	Stage	K_c	p	Critical?
Rice	Initial	1.05	0.20	Yes
	Vegetative	1.10	0.20	No
	Flowering	1.20	0.15	Most
	Grain fill	1.05	0.20	Yes
	Maturity	0.75	—	Stop
Wheat	CRI	0.40	0.40	Most
	Vegetative	0.85	0.55	No
	Flowering	1.15	0.45	Yes
	Grain fill	1.00	0.50	Yes
	Maturity	0.30	—	Stop
Maize	Initial	0.40	0.50	No
	Vegetative	0.85	0.55	No
	Silking	1.20	0.40	Most
	Grain fill	1.05	0.50	Yes
	Maturity	0.60	—	Stop
Tomato	Transplant	0.60	0.30	Yes
	Vegetative	0.90	0.40	No
	Flowering	1.15	0.35	Most
	Fruit bulk	1.15	0.35	Yes
	Maturity	0.80	0.50	Reduce

D. Crop Water Balance (FAO-56)

The total and readily available water in the root zone of depth Z_r are

$$\mathrm{TAW} = \mathrm{AWC} \times Z_r, \tag{10}$$

$$\mathrm{RAW} = p \times \mathrm{TAW}, \tag{11}$$

where p is the stage-specific depletion fraction from FAO-56 Table 22 [7]. Irrigation is triggered when root-zone depletion exceeds RAW. Crop evapotranspiration under standard conditions is

$$ET_c = K_c \times ET_0, \tag{12}$$

with ET_0 obtained from the weather context and K_c from the knowledge base.

Table II summarizes the crop coefficients and depletion fractions for the four core crops, sourced from FAO-56 [7], FAO-33 [8], IRRI Rice Knowledge Bank, and ICAR extension literature. These values populate the Supabase crop˙stages table.

E. Closed-Loop Volumetric Delivery

Once a need is established, the system calculates a target volume rather than running the pump for a fixed duration. Given a net irrigation depth D_{net} and application efficiency η, the target volume for zone area A is

$$V_{\mathrm{target}} = \frac{D_{\mathrm{net}} \times A}{\eta} \quad [\mathrm{L}]. \tag{13}$$

During pumping, the STM32 counts flow-sensor pulses via an interrupt. With calibration factor K_f (pulses per liter), the delivered volume at time t is

$$V_{\mathrm{del}}(t) = \frac{N_{\mathrm{pulses}}(t)}{K_f}. \tag{14}$$

The pump control law is evaluated every interrupt cycle:

$$\text{Pump} = \begin{cases} \text{OFF} & \text{if } V_{\text{del}} \geq V_{\text{target}} \\ \text{OFF} & \text{if } t_{\text{run}} \geq T_{\max} \\ \text{OFF} & \text{if float switch} = \text{LOW} \\ \text{ON} & \text{otherwise.} \end{cases} \tag{15}$$

The first condition closes the volumetric loop; the remaining two are hardware safety overrides.

F. Agro-Environmental Indices

To support interpretability, the backend computes three derived indices from the BME280 readings:

Vapor Pressure Deficit (VPD). Saturation vapor pressure follows Tetens' formula:

$$e_s = 0.6108 \, \exp\!\left(\frac{17.27\, T_a}{T_a + 237.3}\right), \tag{16}$$

yielding $\text{VPD} = e_s\,(1 - RH/100)$ in kPa.

Water Stress Index (WSI). A normalized $[0, 1]$ indicator of root-zone depletion relative to RAW. [TO BE ADDED: Exact formula and threshold mapping.]

Disease Risk Index (DRI). An empirical proxy combining high humidity ($RH > 85\%$), moderate canopy temperature (18–28 °C), and low light to flag fungal-favorable conditions. [TO BE ADDED: Exact scoring logic and validation.]

Every irrigation decision carries one of the following reason codes:

- RUN_MOISTURE_LOW — calibrated moisture below p-threshold.
- RUN_WATER_DEFICIT — high ET_c and accumulated deficit.
- SKIP_MOISTURE_ADEQUATE — soil within safe buffer.
- SKIP_RAIN_CONTEXT — forecast rain covers D_{net}.
- FAULT_TANK_LOW — float switch tripped.
- FAULT_SENSOR_INVALID — ADC out of plausible range.

G. Machine Learning Decision Support

A Random Forest classifier [9] trained with scikit-learn [10] provides an IRRIGATE/NO˙IRRIGATE recommendation as a supplementary signal. The model uses calibrated moisture, soil temperature, BME280 readings, light, crop, and stage as features. It does not control the pump directly; its output is logged alongside the rule-based decision, and disagreements are flagged for review. [TO BE ADDED: Feature importance plot, confusion matrix, and cross-validation metrics from the training run. Model training is complete but evaluation write-up is pending.]

[TO BE ADDED: Close-up photograph of MOSFET driver circuit with gate resistor, pull-down, flyback diode, and pump connector]

Fig. 5. Pump actuation driver circuit of TF01. [TO BE ADDED: Replace with actual photo.]

V. Prototype Implementation and Verification

A. Hardware Assembly

Node TF01 was assembled on a prototype board with three isolated power rails:

- 12 V pump rail (direct from lead-acid battery or adapter).
- 5 V logic rail (LM2596 buck regulator) for the ESP32.
- 3.3 V analog rail (AMS1117-3.3) for the STM32 and sensors.

The pump driver uses an N-channel logic-level MOSFET with a 220 Ω gate resistor, a 10 kΩ pull-down, and a 1N4007 flyback diode across the DC pump terminals. A three-position toggle switch provides hardware-level mode selection: AUTO / OFF / MANUAL.

B. Firmware Verification

The STM32 firmware (embedded C++, STM32 HAL) was tested on the bench over a continuous 48-hour run at a 1 Hz internal polling rate and a 0.1 Hz (10-second) cloud reporting cadence. The following observations were recorded:

1) Telemetry integrity: Zero CRC errors across >10 000 consecutive UART–Wi-Fi–HTTPS transmissions.
2) Kalman noise attenuation: Raw ADC standard deviation $\sigma_{\text{raw}} = 14.8$ counts was reduced to $\sigma_{\text{filt}} = 1.9$ counts after filtering ($Q = 0.01$, $R = 2.5$), an 87% reduction.
3) Wi-Fi resilience: Intentional router power-off triggered local ring-buffer caching on the ESP32; upon reconnection, all buffered packets were delivered without interrupting STM32 execution.
4) Float-switch safety: Triggering the float switch de-energized the pump GPIO within <15 ms, asserting FAULT_TANK_LOW.

[TO BE ADDED: Oscilloscope screenshot or plot showing raw vs. Kalman-filtered ADC trace during bench test.]

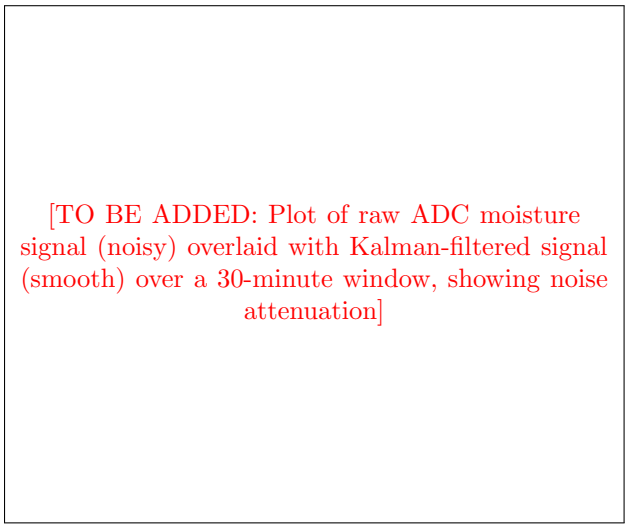


Fig. 6. Raw vs. Kalman-filtered soil-moisture ADC signal. [TO BE ADDED: Replace with actual plot from bench data.]

[TO BE ADDED: Table of sensor readings from a representative 24-hour bench capture.]

VI. Proposed Experimental Protocol

The following experiments are defined for the ongoing Phase 3–4 field deployment. Results will be incorporated once the trials are complete.

A. Experiment 1: Soil-Specific Sensor Calibration

Objective: Establish ADC-to-θ_v transfer functions for each soil class.

Five moisture states (oven-dry, wilting point, half-depletion, field capacity, saturated) will be prepared per soil type. ADC readings will be logged at 10-second intervals for 15 minutes per state. The calibration curve will be fitted using a second-order polynomial:

$$\theta_v = a\,(\mathrm{ADC})^2 + b\,(\mathrm{ADC}) + c, \tag{17}$$

and evaluated by R^2 and RMSE.

[TO BE ADDED: Calibration data, fitted curves, and R^2 values for each soil type.]

B. Experiment 2: Three-Way Controller Comparison

Objective: Quantify TerraFlow water savings and stress prevention against two baselines.

Three plots within Z01 will be instrumented identically:

- Plot A (fixed timer): two daily irrigations, 10 minutes each.
- Plot B (static threshold): pump ON below 40% raw reading, 5-minute run.
- Plot C (TerraFlow): context-aware p-based trigger, pulse-flow volume control, rain-skip.

The primary evaluation metrics are:

$$W_{\text{total}} = \textstyle\sum_{j=1}^{M} V_{\text{del},j}, \tag{18}$$

$$\epsilon_V = \frac{1}{M}\sum_{j=1}^{M} \frac{|V_{\text{target},j} - V_{\text{del},j}|}{V_{\text{target},j}} \times 100\%, \tag{19}$$

$$\text{WS} = \frac{W_{\text{base}} - W_{\text{TF}}}{W_{\text{base}}} \times 100\%. \tag{20}$$

Additional metrics include root-zone stress duration T_{stress} (hours below $p \cdot \text{TAW}$ during critical stages), irrigation event count, and rain-skip utilization rate.

[TO BE ADDED: All experimental results from the three-way comparison.]

C. Experiment 3: Zone Representativeness

A mobile reference probe will be placed at five grid points (1, 3, 5 m from TF01) to measure the spatial coefficient of variation:

$$CV_{\text{zone}} = \frac{\sigma_{\text{zone}}}{\mu_{\text{zone}}} \times 100\%. \tag{21}$$

[TO BE ADDED: Spatial variability data and CV analysis.]

D. Experiment 4: Crop Suitability Validation

[TO BE ADDED: Software validation cases: compatible and incompatible crop-environment scenarios checked against the rule-based scoring engine.]

E. Experiment 5: Fertilizer Advisory Validation

[TO BE ADDED: Soil-test reference cases compared against the TerraFlow recommendation using the documented agronomic method.]

VII. Discussion, Limitations, and Future Scope

The dual-controller topology guarantees that local hardware safety is never subordinated to network availability. This is a deliberate architectural decision: in a rural Indian deployment, Wi-Fi outages are the norm, not the exception. By keeping all pump-critical logic inside the STM32 firmware, TerraFlow avoids the most dangerous failure mode of cloud-first systems.

A. Current Limitations

- Single-zone scope. One node cannot represent a multi-hectare farm. The results from Z01 should not be generalized to heterogeneous terrain without spatial validation.
- TDS $\neq$ NPK. The TDS sensor provides irrigation-water salinity context. Soil nutrient analysis requires certified laboratory testing.
- FAO parameter locality. AWC and K_c values are texture-class averages; site-specific soil samples may shift them.
- ML dataset size. The Random Forest was trained on a limited benchtop dataset. Field data augmentation is needed before claiming generalization.

B. Future Scope

1) LoRaWAN multi-zone mesh. Sub-GHz long-range telemetry would enable Z01–Z10 clustering across kilometer-scale farms.
2) Solar energy autonomy. MPPT charge control with $LiFePO_4$ storage for off-grid deployment.
3) NDVI integration. Multispectral imagery from UAVs or Sentinel-2 to dynamically update K_c values.
4) Digital twin. A real-time simulation mirror for scenario testing and predictive maintenance.

VIII. Conclusion

This paper presented the architecture, mathematical formulation, and prototype verification of TerraFlow BioAI, a zone-aware edge-IoT platform for closed-loop precision irrigation. The key design choices—deterministic STM32 safety authority, on-chip Kalman filtering, FAO-56 volumetric control, and explainable reason codes—address the systemic weaknesses of both fixed-timer schedules and cloud-dependent IoT prototypes. Benchtop verification confirmed reliable multi-sensor telemetry, effective noise attenuation, and sub-15 ms safety-interlock response. A structured field-trial protocol has been defined and is currently being executed; comparative results will be reported upon completion.

Acknowledgment

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